# A CRITERION FOR A HUREWICZ COFIBRATION TO BE A QUILLEN COFIBRATION

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In this paper, we prove that $h$-cofibrations between $q$-cofibrant spaces are $q$-cofibrations and discuss a number of applications. In fact, since our non-equivariant proofs generalise readily to the equivariant context, we work throughout with the $\mathcal{C}$-model structure on $G$-spaces. Here $\mathcal{C}$ is a set of subgroups of $G$ which we take to be closed under conjugacy, and the $\mathcal{C}$-model structure is that of [Sch18, Proposition B.7]. Recall that the $\mathcal{C}$-model structure is a topological model structure cofibrantly generated by:

$$I = \{G/H \times S^{n-1} \to G/H \times D^n | n \geq 0, H \in \mathcal{C}\},$$
$$J = \{G/H \times D^n \to G/H \times D^n \times I | n \geq 0, H \in \mathcal{C}\}$$

Moreover, the $\mathcal{C}$-fibrations (resp. $\mathcal{C}$-equivalences) are the maps $p$ such that $p^H$ is a $q$-fibration (resp. $q$-equivalence) for every $H \in \mathcal{C}$. Note that our assumption that $\mathcal{C}$ is closed under conjugacy results in no loss of generality, and is convenient for some of the proofs below. For example, if $H$ is a subgroup of $G$ and we define $\mathcal{C} \cap H = \{K \cap H | K \in \mathcal{C}\}$, we then have:

**Lemma 1:** *If $X$ is $\mathcal{C}$-cofibrant as a $G$-space, then $X$ is $(\mathcal{C} \cap H)$-cofibrant as an $H$-space.*

## Hurewicz cofibrations between q-cofibrant spaces are q-cofibrations

We now move onto the proof of our main theorem. The key point is the following generalisation of [MP12, Proposition 17.1.4], which admits the same proof:

**Lemma 2:** *Let $i : A \to B$ be an $h$-cofibration and $p : X \to Y$ be a map which has the RLP wrt $B \to B \times I$. Suppose that we are given a commutative square:*

$$\begin{array}{ccc} A & \xrightarrow{v} & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle p} \\ B & \xrightarrow[w]{} & Y \end{array}$$

*and a map $\sigma : B \to X$ such that $p\sigma i = pv$, $\sigma i \simeq v$ over $Y$ and $p\sigma \simeq w$ rel $A$. Then $\sigma$ is homotopic to some $\tau$ with $\tau i = v$ and $p\tau = w$.*

*Proof.* Since $i$ is an $h$-cofibration, $\sigma$ is homotopic to some $\sigma^{'}$ with $\sigma^{'} i = v$ in such a way that the composite homotopy $p\sigma^{'} \simeq w$ is still relative to $A$. So we have reduced to the case where $\sigma i = v$. Let $(H, \lambda)$ represent $(B, A)$ as an $NDR$-pair. Let $K : B \times I \to Y$ be a homotopy between $p\sigma$ and $w$, relative to $A$ and normalised so that $K(b, t) = K(b, 1)$ for $t \geq \lambda(b)$. Since $p$ has the RLP wrt $B \to B \times I$, we can lift this homotopy to $\tilde{K} : B \times I \to X$ starting at $\sigma$. If we define $\tilde{\sigma}(b) = \tilde{K}(b, \lambda(b))$ we solve the lifting problem as desired. □

**Lemma 3:** *An $h$-cofibration between $\mathcal{C}$-cofibrant spaces is a $\mathcal{C}$-cofibration.*

*Proof.* Let $i : A \to B$ be an $h$-cofibration between $\mathcal{C}$-cofibrant spaces. We can factor $i$ as a $\mathcal{C}$-cofibration followed by a $\mathcal{C}$-acyclic $\mathcal{C}$-fibration between $\mathcal{C}$-cofibrant objects, so by the retract argument it suffices to show that $i$ has the LLP wrt $\mathcal{C}$-acyclic $\mathcal{C}$-fibrations between $\mathcal{C}$-cofibrant objects.

If $p : X \to Y$ is a $\mathcal{C}$-acyclic $\mathcal{C}$-fibration between $\mathcal{C}$-cofibrant objects, then there exists $s : Y \to X$ such that $ps = 1$ and $sp \simeq 1$ over $Y$. Since $B$ is $\mathcal{C}$-cofibrant $p$ has the RLP wrt $B \to B \times I$. Therefore, we can solve a lifting problem as in Lemma 2, with the initial up to homotopy lift $\sigma$ as $sw$. □

**Example 4:** *Suppose that $H \in \mathcal{C}$. If $U$ is an equivariant open subspace of $G/H \times D^n$, then $i : U \cap (G/H \times \partial D^n) \to U \cap (G/H \times D^n)$ is the inclusion of the boundary of a smooth $G$-manifold and so is an $h$-cofibration between $\mathcal{C}$-cofibrant spaces by the equivariant collar neighbourhood theorem, [Kan07, Theorem 3.5]. Therefore, Lemma 3 tells us that $i$ is a $\mathcal{C}$-cofibration.*

**Lemma 5:** *If $i : A \to B$ is a $\mathcal{C}$-cofibration and $U \subset B$ is an equivariant open subspace, then $i : A \cap U \to U$ is a $\mathcal{C}$-cofibration.*

*Proof.* Firstly, $i$ is a retract of a relative $\mathcal{C}$-cell complex, $j : A \to C$, as shown below:

$$\begin{tikzcd} A \arrow[r] \arrow[d, "i"] & A \arrow[r] \arrow[d, "j"] & A \arrow[d, "i"] \\ B \arrow[r, "i"'] & C \arrow[r, "r"'] & B \end{tikzcd}$$

If we let $V = r^{-1}(U)$, then $i : A \cap U \to U$ is a retract of $j : A \cap V \to V$. So we have reduced to the case where $i$ is a relative $\mathcal{C}$-cell complex, which we now assume.

We can express $i$ as a transfinite composite of maps $A_\lambda \to A_{\lambda+1}$, each of which is a pushout of a map of the form $G/H \times S^{n-1} \to G/H \times D^n$, with $H \in \mathcal{C}$. By [Ron23, Lemma 6.2.3] and Lemma 18, the map $i : A \cap U \to U$ can be expressed as a transfinite composite of the maps $A_\lambda \cap U \to A_{\lambda+1} \cap U$, each of which is a pushout of a map of the form $(G/H \times S^{n-1}) \cap \alpha^{-1}(U) \to (G/H \times D^n) \cap \alpha^{-1}(U)$, where $\alpha$ is the inclusion of the cell into $B$. By Example 4, $A_\lambda \cap U \to A_{\lambda+1} \cap U$ is therefore a $\mathcal{C}$-cofibration, and, hence, so is the transfinite composite $i : A \cap U \to U$. □

**Corollary 6:** *An equivariant open subspace of a $\mathcal{C}$-cofibrant $G$-space is $\mathcal{C}$-cofibrant.*

Combining Lemma 3 and Corollary 6, we can rephrase Lemma 3 as follows:

**Theorem 7:** *If $i : A \to B$ is an $h$-cofibration and $B$ is $\mathcal{C}$-cofibrant, then $A$ is $\mathcal{C}$-cofibrant and $i$ is a $\mathcal{C}$-cofibration.*

*Proof.* If $(H, \lambda)$ represents $(B, A)$ as a $G$-NDR-pair, then $A$ is a retract of $\lambda^{-1}([0, 1))$ which is $\mathcal{C}$-cofibrant by Corollary 6. □

**Example 8:** *Let $G = \Sigma_n$ and let $\mathcal{C}$ be the set of subgroups of $G$, each isomorphic to a product of symmetric groups $\Sigma_{n_1} \times ... \times \Sigma_{n_l}$, corresponding to partitions of $\{1, ..., n\}$ into sets of size $n_1, ..., n_l$. Let $i : S^{k-1} \to D^k$ be the boundary inclusion. If we consider the iterated pushout product map $j : S^{k-1} \times (D^k)^{n-1} \cup ... \cup (D^k)^{n-1} \times S^{k-1} := Q_n(i) \to (D^k)^n$, then $j$ is an $h$-cofibration by [May72, Lemma A.4], and the domain and codomain are $\mathcal{C}$-cofibrant since they are $G$-manifolds, hence $G$-CW complexes, [Ill83, Corollary 7.2], with the correct isotropy groups. Therefore, Lemma* **??** *tells us that $j$ is a $\mathcal{C}$-cofibration.*

We then have the following application to the theory of symmetrizable cofibrations, which is a slight generalisation of [Sch18, Proposition 2.1.12]:

**Lemma 9:** *Let $j : A \to B$ be a symmetrizable (acyclic) cofibration in a tensored and cotensored closed symmetric monoidal topological model category. Then for every $k \geq 0$, the pushout product map:*

$$j \Box i : A \times D^k \cup_{A \times S^{k-1}} B \times S^{k-1} \to B \times D^k$$

*is a symmetrizable (acyclic) cofibration.*

*Proof.* For any $n \geq 1$, we want to show that $k := (j \Box i)^{\Box n}/\Sigma_n$ is an (acyclic) cofibration. Equivalently, we want to show that $k$ has the $\Sigma_n$-equivariant LLP with respect to (acyclic) fibrations $p : X \to Y$, where $\Sigma_n$ acts trivially on $X$ and $Y$. Using the tensor-cotensor adjunction, it suffices to show that $i^{\Box n}$ has the equivariant LLP wrt $p^{\Box j^{\Box n}}$. In Example 8, we saw that $i^{\Box n}$ is a $\mathcal{C}$-cofibration, with respect to the specified set of subgroups $\mathcal{C}$. So it suffices to show that $p^{\Box j^{\Box n}}$ is a $\mathcal{C}$-acyclic $\mathcal{C}$-fibration. If $H \in \mathcal{C}$, this amounts to showing $p^{\Box j^{\Box n}/H}$ is a $q$-acyclic $q$-fibration. This follows from the fact that the model structure is topological and monoidal, $j$ is a symmetrizable cofibration and either $p$ is acyclic or $j$ is a symmetrizable acyclic cofibration. □

**Other Examples**

**Definition 10:** *A numerable open cover $\{U_j\}_{j\in J}$ of a $G$-space, $B$, is a locally finite open cover of $B$ such that for every $j \in J$ there exists an equivariant map $\lambda_j : B \to I$ with $U_j = \lambda_j^{-1}((0,1])$.*

**Lemma 11:** *Suppose that $\{U_j\}_{j\in J}$ is a numerable open cover of $B$. If $f : A \to B$ is a map such that $f : f^{-1}(U_j) \to U_j$ is an $h$-cofibration for all $j \in J$, then $f : A \to B$ is an $h$-cofibration.*

*Proof.* [Dol68, Satz 2]. □

The next result is the equivariant analogue of [DE72, Corollary III.2].

**Lemma 12:** *Suppose that $\mathcal{C}$ is closed under conjugacy and intersections. If $X$ is $\mathcal{C}$-cofibrant, then $\Delta : X \to X \times X$ is a $\mathcal{C}$-cofibration.*

*Proof.* The condition that $\mathcal{C}$ is closed under conjugacy and intersections ensures that $X\times X$ is also $\mathcal{C}$-cofibrant. Since any $\mathcal{C}$-cofibrant $G$-space is a retract of a $\mathcal{C}$-cell complex, by Lemma 3 it suffices to show that if $X$ is a $G$-cell complex then $\Delta : X \to X \times X$ is an $h$-cofibration. To this end, assume that $\Delta : Y \to Y \times Y$ is an $h$-cofibration and $X$ is obtained from $Y$ by attaching a single cell along $\alpha : G/H \times S^{n-1} \to Y$. We will show that if $(H, \lambda)$ represents $(Y \times Y, Y)$ an NDR-pair, then there is an NDR pair $(K, \mu)$ for $(X \times X, X)$ such that $K|_{Y\times Y\times I} = H$ and $\mu|_{Y\times Y} = \lambda$. Firstly, the composite $Y \to Y \times Y \to Y \times X \cup_{Y\times Y} X \times Y$ is an $h$-cofibration, represented by an NDR pair that extends $(H, \lambda)$, where note $Y \times Y \to Y \times X \cup_{Y\times Y} X \times Y$ is an $h$-cofibration by [Str72, Lemma 5]. We then consider the map between pushouts:

$$\begin{tikzcd}
G/H \times D^n \arrow[d, "\Delta"] & G/H \times S^{n-1} \arrow[l] \arrow[r] \arrow[d, "\Delta"] & Y \arrow[d, tail, "\alpha"] \\
G/H \times G/H \times D^n \times D^n & G/H \times G/H \times \partial D^{2n} \arrow[l] \arrow[r] & Y \times X \cup_{Y\times Y} X \times Y
\end{tikzcd}$$

We've seen that $\alpha$ is an $h$-cofibration. The map between pushouts is $\Delta : X \to X \times X$ and to show this is an $h$-cofibration it suffices to show the pushout-product map, $\phi$, below is an $h$-cofibration:

$$\phi : G/H \times D^n \cup_{G/H\times S^{n-1}} G/H \times G/H \times \partial D^{2n} \to G/H \times G/H \times D^n \times D^n$$

Let $U \cong G/H \times \partial D^n \times [0, 1)$ be a numerable boundary collar in $G/H \times D^n$. Then consider the restriction of $\phi$ to $U \times U$, which can be identified with:

$$G/H\times\partial D^n\times[0,1)\cup G/H\times G/H\times\partial D^n\times\partial D^n\times\partial([0,1)\times[0,1)) \to G/H\times G/H\times\partial D^n\times\partial D^n\times[0,1)\times[0,1)$$

Let $L \subset [0, 1) \times [0, 1)$ be the union of the diagonal and the boundaries, so $L$ consists of three straight lines. The map above is the composite of the pushout product of $\Delta : G/H \times \partial D^n \to G/H \times G/H \times \partial D^n \times \partial D^n$ and $\partial([0, 1] \times [0, 1)) \to L$ with the map:

$$G/H \times G/H \times \partial D^n \times \partial D^n \times L \to G/H \times G/H \times \partial D^n \times \partial D^n \times [0, 1) \times [0, 1)$$

and is therefore an $h$-cofibration as desired. If $V$ is the complement in $G/H\times D^n$ of $G/H\times\partial D^n\times[0, 1/2] \subset U$, then $V$ is also numerable and the restriction of $\phi$ to each of $U \times U$, $V \times G/H \times D^n$ and $G/H \times D^n \times V$ is an $h$-cofibration. Therefore, $\phi$ is an $h$-cofibration by [Dol68, Satz 2], and, therefore, $\Delta : X \to X \times X$ is an $h$-cofibration, since it is the composite of a pushout of $\alpha$ with a pushout of $\phi$. Etc... □

As in [DE72], we make the following definition:

**Definition 13:** *We call a $G$-space, $X$, $G$-equivariantly locally equiconnected (or $G$-LEC) if $\Delta : X \to X \times X$ is an $hG$-cofibration.*

The next lemma is the equivariant analogue of [DE72, Theorem II.3]:

**Lemma 14:** *Let $X$ be $G$-LEC and suppose that $x \in X$ has isotropy group $H$. Then there exists an $H$-equivariant open neighbourhood $U$ of $x$ such that $\pi_1, \pi_2 : U \times U \to X$ are $H$-homotopic. Here, $\pi_i$ denotes the projection onto the $i$th factor followed by the inclusion into $X$.*

*Proof.* Let $(H, \lambda)$ represent the inclusion of the diagonal $(X \times X, \Delta)$ as a $G$-NDR pair. Identify $X$ with $\{x\} \times X$ and let $U = \lambda^{-1}([0,1)) \cap (\{x\} \times X)$. Consider $H : \{x\} \times U \times I \to X \times X$. Then $\pi_2 H$ defines an $H$-homotopy from the inclusion $i : U \to X$ to some map $f$. Moreover, $\pi_1 H$ defines an $H$-homotopy from $f$ to the constant map to $x$. If we now consider $\pi_1, \pi_2 : U \times U \to X$, we see that $\pi_1$ is the composite $U \times U \to X \times U \xrightarrow{\pi_1} X$, which is $H$-homotopic to the constant map to $x$. Similarly, $\pi_2$ is $H$-homotopic to the constant map to $x$, so $\pi_1$ and $\pi_2$ are $H$-homotopic. □

**Lemma 15:** *Let $X$ be $\mathcal{C}$-cofibrant and suppose that $x \in X$ has isotropy group $H$. Then there exists an $H$-equivariant open neighbourhood $U$ of $x$ such that $\pi_1, \pi_2 : U \times U \to X$ are $H$-homotopic via an $H$-homotopy which is constant on $\Delta$.*

*Proof.* By Lemma 12 and Lemma 14, we know that $\pi_1$ and $\pi_2$ are $H$-homotopic, say via an $H$-homotopy $L : U \times U \times I \to X$. Define $\Gamma : \Delta \times I \to X$ to be the restriction of $L$ to $\Delta \times I$. Let $\Gamma^{-1}$ denote the inverse homotopy from $\pi_2$ to $\pi_1$. Define $K$ to be composite (via concatenation) homotopy of $L$ and $\Gamma^{-1}\pi_2$:

$$U \times U \times I \xrightarrow{\pi_2} U \times I \cong \Delta \times I \xrightarrow{\Gamma^{-1}} X$$

Then the restriction of $K$ to $\Delta \times I$ is the composite of $\Gamma$ and $\Gamma^{-1}$, and $K$ is an $H$-homotopy from $\pi_1$ to $\pi_2$. The composite of $\Gamma$ and $\Gamma^{-1}$ is itself $H$-homotopic, say via $\alpha : \Delta \times I \times I \to X$, to the constant homotopy on $\pi_1|_\Delta$, and $\alpha$ can be made relative to $\Delta \times \{0,1\} \times I$. We can then define a lifting problem:

$$\begin{tikzcd}
U \times U \times I \times \{0\} \cup \Delta \times I \times I \cup U \times U \times \{0,1\} \times I \arrow[r, "K \cup \alpha \cup cnst(\{\pi_1, \pi_2\})"] \arrow[d] & X \arrow[d] \\
U \times U \times I \times I \arrow[r] & *
\end{tikzcd}$$

A lift $\tilde{K} : U \times U \times I \times I \to X$ exists since $U$ is $(\mathcal{C} \cap H)$-cofibrant, by Corollary 6 and Lemma 1, and, therefore, $H$-LEC by Lemma 12. This implies the left hand vertical map is an $hH$-acyclic $hH$-cofibration. Restricting $\tilde{K}$ to $U \times U \times I \times \{1\}$ gives the desired $H$-homotopy. □

We will use the following consequence of [Las82, Lemma 2.8], which was noted in [BJ15, Lemma 6.1]:

**Lemma 16:** *If $p : E \to B$ is an $hH$-fibration, then $1 \times_H p : G \times_H E \to G \times_H B$ is an $hG$-fibration.*

*Proof.* Consider a lifting problem of $G$-spaces as in the right hand square below:

$$\begin{tikzcd}
G \times_H V_0 \arrow[r, "\cong"] \arrow[d] & X \arrow[r, "f"] \arrow[d] & G \times_H E \arrow[d] \\
G \times_H (V_0 \times I) \arrow[r, "\cong"'] & X \times I \arrow[r, "\alpha"'] & G \times_H B
\end{tikzcd}$$

Then $X \times I \cong G \times_H \alpha^{-1}(B)$ and so [Las82, Lemma 2.8] tells us that there is an $H$-homeomorphism $\psi : V_0 \times I \to V$, where $V = \alpha^{-1}(B)$, $V_0 = V \cap (X \times \{0\})$ and $\psi|_{V_0 \times \{0\}}$ is the inclusion of $V_0$ into $V$. Note that $G \times_H V_0 \cong X$. The $H$-homeomorphism, $\psi$, induces a $G$-homeomorphism $G \times_H (V_0 \times I) \to X \times I$, as in the diagram above, which restricts at $0 \in I$ to a homeomorphism onto $X \times \{0\}$. So to solve the lifting problem of the right hand square, it suffices to solve the adjoint lifting problem of $H$-spaces:

$$\begin{tikzcd}
V_0 \arrow[r, "f"] \arrow[d] & G \times_H E \arrow[d] \\
V_0 \times I \arrow[r, "\alpha\psi"'] & G \times_H B
\end{tikzcd}$$

Since $V = \alpha^{-1}(B)$, $\alpha\psi : V_0 \times I \to G \times_H B$ factors through $B$, and $f : V_0 \to G \times_H E$ factors through $E$. Since $p$ is an $hH$-fibration, it follows that the lifting problem can be solved. □

A version of the following theorem was first proved in an equivariant context by Gevorgyan and Jimenez in [GJ19, Theorem 6], using the $qG$-model structure and under the assumption that both $E$ and $B$ are $G$-CW complexes. The proof below corrects a minor gap in that proof, since it is not necessarily true that a $G$-CW complex, $X$, can be covered by $G$-equivariant open subspaces which are non-equivariantly contractible in $X$.

**Theorem 17:** *Suppose that $\mathcal{C}$ is closed under conjugacy and intersections. If $p : E \to B$ is a $\mathcal{C}$-fibration between $\mathcal{C}$-cofibrant spaces, then $p$ is an $hG$-fibration.*

*Proof.* Let $(H, \lambda)$ represent $(B \times B, \Delta_B)$ as a $G$-NDR pair. As in Lemma 15, if $b \in B$ has isotropy group $H$ there is an $H$-equivariant open neighbourhood $U$ of $b$ and an $H$-homotopy $K : U \times U \times I \to B$ between $\pi_1$ and $\pi_2$ such that $K(u, v, t) = K(u, v, 1)$ for all $t \geq \lambda(u, v)$. Note that $H \in \mathcal{C}$ since there exists an inclusion of $B$ into a $\mathcal{C}$-cell complex. Since $p : E \to B$ is a $(\mathcal{C} \cap H)$-fibration, $\mathcal{C}$ is closed under intersections and both $U$ and $p^{-1}(U)$ are $(\mathcal{C} \cap H)$-cofibrant, we can solve the lifting problem:

$$\begin{tikzcd}
U \times p^{-1}(U) \arrow[r, "\pi_2"] \arrow[d] & E \arrow[d, "p"] \\
U \times p^{-1}(U) \times I \arrow[r, "K\circ(1\times p\times 1)"'] & B
\end{tikzcd}$$

to define a slicing function $\tilde{K} : U \times p^{-1}(U) \times I \to E$. We now show that $p : p^{-1}(U) \to U$ is an $hH$-fibration. Indeed, given a lifting problem of $H$-spaces:

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d] & p^{-1}(U) \arrow[d, "p"] \\
X \times I \arrow[r, "L"'] & U
\end{tikzcd}$$

a lift can be defined via $\tilde{L}(x, t) = \tilde{K}(L(x, t), f(x), \lambda(L(x, t), pf(x)))$.

Since $B$ is completely regular, $b$ also has a $G$-equivariant open neighbourhood $G$-homeomorphic to $G \times_H S$, where $S$ is an $H$-equivariant subspace of $B$ containing $b$, by [Bre72, Theorem 5.4]. Replacing $S$ by $S \cap U$ if necessary, we may as well assume $S \subset U$. In this case, it follows from the above that $p : p^{-1}(S) \to S$ is an $hH$-fibration and so $p : G \times_H p^{-1}(S) \to G \times_H S$ is an $hG$-fibration by Lemma 16. It follows that for every $b \in B$ there is a $G$-equivariant neighbourhood, $V$, of $b$ for which $p : p^{-1}(V) \to V$ is an $hG$-fibration. Since $B$ is paracompact, it follows that $p$ is an $hG$-fibration. □

**Appendix of point set topology**

**Lemma 18:** *Let $I$ be a filtered category and let $D : I \to \mathcal{U}$ be a functor which takes maps in $I$ to closed inclusions. Let $A := colimD$ and $f : X \to A$ be a map. Define $Y(i)$ to be the pullback of $D(i) \to A$ along $f$. Then the canonical map $colimY \to X$ is a homeomorphism.*

*Proof.* We can express filtered colimits of closed inclusions, such as $A$, as quotients of the disjoint union of the colimiting subspaces, $\sqcup_{i\in I} D(i) \to A$, [Str09, Lemma 3.3]. We have a pullback:

$$\begin{tikzcd}
\sqcup_{i\in I} Y_i \arrow[r] \arrow[d] & X \arrow[d, "f"] \\
\sqcup_{i\in I} D_i \arrow[r] & A
\end{tikzcd}$$

Since the pullback of a quotient map is a quotient map, [Str09, Propostion 2.36], we obtain the result. □

**Lemma 19:** *The following properties of a space $X$ are closed under retracts: being i) Hausdorff, ii) paracompact, iii) completely regular.*

*Proof.* Suppose that we have maps $i : A \to X, r : X \to A$ with $ri = 1$. If $X$ is Hausdorff, let $a$ and $b$ be distinct points in $A$. Then there exists disjoint open sets $U$ and $V$ in $X$ separating $i(a)$ and $i(b)$. So $i^{-1}(U)$ and $i^{-1}(V)$ are disjoint and separate $a$ and $b$. So $A$ is Hausdorff.

If $X$ is paracompact and $\{U_i\}_{i \in I}$ is an open cover of $A$, then $\{r^{-1}(U_i)\}$ is an open cover of $X$ which admits a locally finite open refinement $\{V_j\}_{j \in J}$. Then, for each $j$ $i^{-1}(V_j)$ is contained within $i^{-1}r^{-1}(U_i) = U_i$ for some $i$, so $\{i^{-1}(V_j)\}$ is an open refinement of $\{U_i\}$. Moreover, for every $a \in A$ there is an open neighbourhood, $W$, of $i(a)$ which intersects only finitely many of the $V_j$. Then $i^{-1}(W)$ is an open neighbourhood of $a$, intersecting only finitely many of the $i^{-1}(V_j)$. So $A$ is paracompact.

Next suppose that $X$ is completely regular. Let $a \in A$ and $C$ be a closed subspace of $A$ not containing $a$. Then $r^{-1}(C)$ is a closed subspace of $X$ not containing $i(a)$. So there exists a continuous function $\lambda : X \to [0, 1]$ such that $\lambda i(a) = 0$ and $\lambda(r^{-1}(C)) = 1 \implies \lambda i(C) = 1$. So $A$ is completely regular. □

## References


[BJ15] Alexander Bykov and Raúl Juárez Flores. G-fibrations and twisted products. *Topology and its Applications*, 196:379–397, 2015. Proceedings of the International Conference on Topology and Geometry 2013, joint with the Sixth Japan-Mexico Topology Symposium (September 2-6, 2013, Matsue, Japan).

[Bre72] G. Bredon. *Introduction to Compact Transformation Groups*, volume 46 of *Pure annd Applied Mathematics*. Elsevier, 1972.

[DE72] Eldon Dyer and S. Eilenberg. An adjunction theorem for locally equiconnected spaces. *Pacific Journal of Mathematics*, 41(3):669 – 685, 1972.

[Dol68] A. Dold. Die homotopieerweiterungseigenschaft (=hep) ist eine lokale eigenschaft. *Inventiones mathematicae*, 6(3):185–189, 1968.

[GJ19] P. S. Gevorgyan and R. Jimenez. On equivariant fibrations of g-cw-complexes. *Sbornik: Mathematics*, 210(10):1428, oct 2019.

[Ill83] S. Illman. The equivariant triangulation theorem for actions of compact lie groups. *Mathematische Annalen*, 262:487–501, 1983.

[Kan07] Marja Kankaanrinta. Equivariant collaring, tubular neighbourhood and gluing theorems for proper Lie group actions. *Algebr. Geom. Topol.*, 7:1–27, 2007.

[Las82] R. K. Lashof. Equivariant bundles. *Illinois Journal of Mathematics*, 26(2):257 – 271, 1982.

[May72] J.P. May. *The Geometry of Iterated Loop Spaces*. Lecture Notes in Mathematics. Springer Berlin, Heidelberg, 1972.

[MP12] J. P. May and K. Ponto. *More Concise Algebraic Topology: localization, completion, and model categories*. Chicago lectures in mathematics. University of Chicago Press, 2012.

[Ron23] A. Ronan. *Nilpotent Groups, Spaces and G-Spaces*. PhD thesis, University of Warwick, 2023. `https://wrap.warwick.ac.uk/id/eprint/187006/1/WRAP_Theses_Ronan_2023.pdf`.

[Sch18] Stefan Schwede. *Global Homotopy Theory*. New Mathematical Monographs. Cambridge University Press, 2018.

[Str72] A. Strøm. The homotopy category is a homotopy category. *Arch. Math.*, 23:435–441, 1972.

[Str09] Neil P. Strickland. The category of CGWH spaces. 2009. https://ncatlab.org/nlab/files/StricklandCGHWSpaces.pdf.